e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
( Peer-Reviewed, Open Access, Fully Refereed International Journal )
Volume:08/Issue:03/March-2026 Impact Factor- 8.187 www.irjmets.com

# TRANSFER LEARNING FRAMEWORK FOR PREDICTION OF ALZHEIMER'S DISEASE

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DOI: https://www.doi.org/10.56726/IRJMETS91989

## ABSTRACT

Alzheimer's disease is a progressive neurological disorder that impacts memory, thinking skills, and overall cognitive function. Catching it early is crucial for slowing its progression and aiding in clinical decision-making. This research introduces a deep learning approach to predict Alzheimer's disease using MRI brain images and transfer learning models. We implemented and evaluated several pretrained architectures, including VGG16, ResNet50, MobileNetV2, and EfficientNetB0. To enhance generalization, the MRI dataset underwent preprocessing, which included resizing, normalization, and augmentation techniques. Initially, we trained and assessed all models based on their validation accuracy. Among them, EfficientNetB0 stood out with a validation accuracy of 79.27%. We further refined the model through hyperparameter tuning, adjusting the learning rate, dropout, and the number of dense layer units. The optimal performance was achieved with a learning rate of 0.0001, a dropout of 0.4, and 128 dense units. This tuned model reached an impressive validation accuracy of 95.14%, with enhanced precision and recall across all Alzheimer's classes. These results highlight that our transfer learning approach can provide accurate and reliable predictions of Alzheimer's disease from MRI images, aiding in early diagnosis.

**Keywords:** Alzheimer's Disease, Transfer Learning, Deep Learning, EfficientNetB0, MRI Classification, Hyperparameter Tuning.

## I. INTRODUCTION

Alzheimer's disease is a progressive neurological disorder that impacts memory, thinking skills, and cognitive behavior. It stands as one of the leading causes of dementia in older adults. As the disease progresses, it gradually damages brain cells, resulting in memory loss and challenges in carrying out everyday tasks. Detecting Alzheimer's early on is crucial for slowing its progression and enhancing patient care.

Magnetic Resonance Imaging (MRI) is commonly utilized to examine structural changes in the brain. The images produced by MRI offer detailed insights into brain tissues and any abnormalities present. However, analyzing these MRI images manually can be quite time-consuming and relies heavily on expert radiologists. This is where automated prediction systems powered by deep learning techniques come into play.

Recent strides in deep learning have demonstrated impressive results in the classification of medical images. Transfer learning models like VGG16, ResNet50, MobileNetV2, and EfficientNetB0 are frequently employed for image classification tasks. These pretrained models not only cut down on training time but also enhance accuracy. In this research, several transfer learning models are applied to predict Alzheimer's disease, with the top-performing model being fine-tuned through hyperparameter adjustments to achieve even better accuracy.

## II. METHODOLOGY

The methodology we're proposing is all about predicting Alzheimer's disease through deep learning and transfer learning models. The process involves several key steps: first, we preprocess the dataset, then we train the model, fine-tune the hyperparameters, and finally, we evaluate the results.

### Dataset Collection

We gather MRI brain images and sort them into four categories: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. This dataset is then split into training and validation sets.

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**Data Preprocessing**

All images are resized to 224 × 224 pixels, and we normalize the pixel values to fall between 0 and 1. To enhance the model's ability to generalize, we apply data augmentation techniques like rotation, flipping, and zooming.

**Transfer Learning Models**

We utilize pretrained models such as VGG16, ResNet50, MobileNetV2, and EfficientNetB0, all loaded with ImageNet weights. We also add fully connected layers specifically for classifying Alzheimer's disease.

**Training Process**

During training, we use the categorical cross entropy loss function along with the Adam optimizer. We keep a close eye on both training accuracy and validation accuracy throughout the process.

**Hyperparameter Tuning**

To boost the model's performance, we fine-tune parameters like the learning rate, dropout rate, and the number of units in the dense layer. The best settings are chosen based on the validation accuracy.

## III. MODELING AND ANALYSIS

In this project, we explored various transfer learning models to predict Alzheimer's disease.

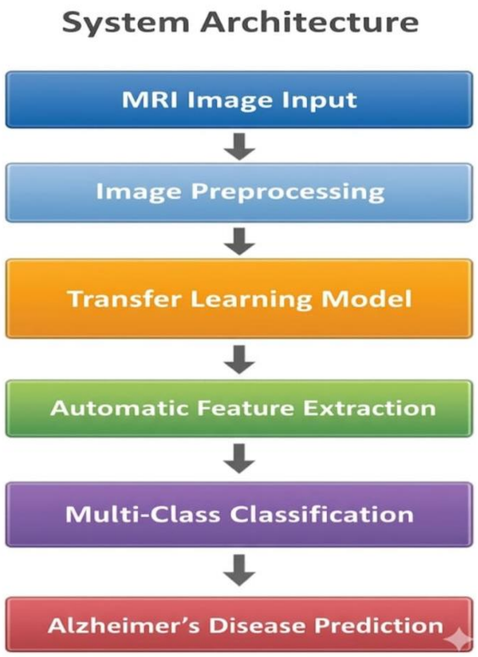


**Figure 1:** Proposed Transfer Learning Architecture for Alzheimer's Disease Prediction

We've set up a few models using pretrained ImageNet weights and fine-tuned them with MRI brain images.

**VGG16 Model**

The VGG16 model is built with deep convolutional layers that excel at feature extraction. We trained it for 5 epochs and assessed its performance using validation accuracy.

**ResNet50 Model**

ResNet50 incorporates residual connections to tackle the vanishing gradient problem. Similar to VGG16, this model was also trained for 5 epochs and evaluated.

**MobileNetV2 Model**

MobileNetV2 stands out with its depth wise separable convolution and inverted residual blocks. This design not only cuts down on computation costs but also boosts performance.

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**EfficientNetB0 Model**

EfficientNetB0 takes a unique approach by scaling depth, width, and resolution to enhance performance. It outperformed the other models we tested, making it a noteworthy contender.

## IV. RESULTS AND DISCUSSION

Initially, we explored several pretrained architectures, including VGG16, ResNet50, MobileNetV2, and EfficientNetB0. We took a close look at how each model performed and compared their results. After evaluating them, we chose the top performer for some fine-tuning of its hyperparameters. We made further enhancements by tweaking the learning rate, dropout, and parameters for the dense layers. To assess the final model's performance, we used a confusion matrix and a classification report.

**Table 1.** Results Before Hyperparameter Tuning

| S.No | Model | Epochs | Best Validation Accuracy |
|---|---|---|---|
| 1 | VGG16 | 5 | 66.03% |
| 2 | ResNet50 | 5 | 52.45% |
| 3 | MobileNetV2 | 5 | 68.36% |
| 4 | EfficientNetB0 (Baseline) | 6 / 15 (Early Stopped) | 79.27% |

As shown in Table 1, EfficientNetB0 stood out with the highest validation accuracy among the models we tested. VGG16 and ResNet50 didn't perform as well, while MobileNetV2 landed somewhere in the middle. Hence, we decided to focus on EfficientNetB0 for our hyperparameter tuning.

For the tuning process, we adjusted the learning rate, dropout, and the number of dense units. We experimented with various combinations to boost performance. The sweet spot turned out to be a learning rate of 0.0001, a dropout value of 0.4, and 128 dense units, which led to an impressive validation accuracy of 95.14%.

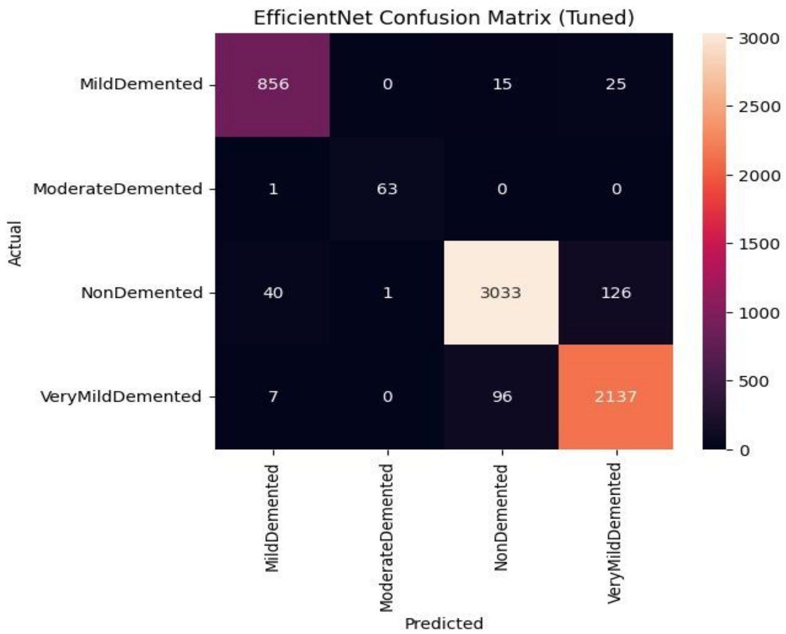


**Figure 2:** Confusion matrix of tuned EfficientNet model

Figure 2 illustrates the confusion matrix for the tuned EfficientNet model, showing that most samples were accurately classified across all four Alzheimer classes. The model performed particularly well in identifying Mild Demented, Moderate Demented, Non Demented, and Very Mild Demented categories.

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| | precision | recall | f1-score | support |
|---|---|---|---|---|
| MildDemented | 0.95 | 0.96 | 0.95 | 896 |
| ModerateDemented | 0.98 | 0.98 | 0.98 | 64 |
| NonDemented | 0.96 | 0.95 | 0.96 | 3200 |
| VeryMildDemented | 0.93 | 0.95 | 0.94 | 2240 |
| accuracy | | | 0.95 | 6400 |
| macro avg | 0.96 | 0.96 | 0.96 | 6400 |
| weighted avg | 0.95 | 0.95 | 0.95 | 6400 |

**Figure 3:** Classification report of tuned EfficientNet model

Figure 3 presents the classification report for the tuned model, highlighting its high precision, recall, and F1-score across all classes. Overall, the model achieved an accuracy of 95%, showcasing its strong classification capabilities.

## V. CONCLUSION

In this project, we dive into predicting Alzheimer's disease using transfer learning models. We've put several pretrained architectures to the test, including VGG16, ResNet50, MobileNetV2, and EfficientNetB0. Among these, EfficientNetB0 stood out with the best performance. By fine-tuning the hyperparameters, we were able to boost the model's accuracy even further. Ultimately, the final tuned model reached an impressive validation accuracy of 95.14%. This proposed system offers a reliable and efficient way to detect Alzheimer's disease early on, utilizing MRI brain images. Looking ahead, we plan to work with a larger dataset and explore ensemble learning techniques to enhance performance even more.

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